

Day 3: Scenario 3 Model Comparison

GSK Medicine Prediction System | One-Page Summary

What was completed

- Readmission conversion fix was retained, and preprocessing remained training-only to avoid leakage.
- A fixed 60/20/20 split (train/validation/test) was used for comparison across five saved models.
- No retraining, no retuning, and no split changes were performed in this summary step.
- LSTM is included as a one-timestep educational demonstration (one row per patient).

Test results: Treatment outcome prediction

Precision, recall and F1 refer to Effective (1).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	57.95%	29.97%	66.03%	0.4123	0.6459
Random Forest	58.02%	29.90%	65.41%	0.4104	0.6449
XGBoost	72.39%	36.37%	31.54%	0.3379	0.6488
Neural Network	59.68%	30.48%	62.85%	0.4105	0.6496
LSTM Demo	59.27%	30.37%	63.73%	0.4114	0.6500

Key findings

- Neural Network was selected using validation ROC-AUC among the four eligible models; the top eligible ROC-AUC gap is small (0.0014).
- XGBoost has the highest test accuracy (72.39%) and precision (36.37%).
- The 50% precision target remains unmet (best test precision: 36.37%).
- Always-zero baseline accuracy is 77.67% with zero recall.
- The test split was also evaluated in earlier scenarios.

This page is generated from outputs/scenario3_final_model_comparison.csv and outputs/scenario3_final_model_comparison_report.txt.